

Unit III – Ideation and Creative Thinking

Detailed lecture notes — idea generation techniques, brainstorming and brainwriting, mind mapping, SCAMPER, lateral thinking and the Six Thinking Hats, and the structured selection of concepts.

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SYLLABUS

UNIT III — IDEATION AND CREATIVE THINKING (6 periods)

Idea generation techniques. Brainstorming. Mind mapping. SCAMPER technique. Lateral thinking. Concept selection methods. Feasibility and innovation evaluation. Team ideation exercises.

Course Outcome CO3: Generate and evaluate innovative ideas using structured ideation techniques.

Text: Tim Brown, *Change by Design*, Harper Collins, 2012; Idris Mootee, *Design Thinking for Strategic Innovation*, 1st Edn., John Wiley & Sons, 2013.

References: Alex F. Osborn, *Applied Imagination*, 1953; Edward de Bono, *Six Thinking Hats*, 1985; Stuart Pugh, *Total Design*, Addison-Wesley, 1990; Nigel Cross, *Engineering Design Methods*, 4th Edn., Wiley, 2008.

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Learning objectives. After studying this unit you should be able to: (i) explain the purpose of ideation and why quantity is pursued before quality; (ii) select an appropriate ideation technique by medium and by degree of structure, and plan and facilitate a session; (iii) state Osborn's rules of

brainstorming and explain the causes of productivity loss in interacting groups; (iv) run a 6-3-5 brainwriting session and describe three other written methods; (v) construct a mind map and convert it into named themes; (vi) apply all seven SCAMPER prompts to an existing product; (vii) distinguish vertical from lateral thinking, use provocation and random entry, and run a Six Thinking Hats session; and (viii) converge on concepts using clustering, dot voting, the three-lens screen and a Pugh matrix, and identify the common traps in selection.

1. Introduction – what ideation is for

Unit II ended with a point-of-view statement and a set of How Might We questions. Ideation is the deliberate, time-boxed search for options that answer one of those questions. It is not a debate about which option to take; judgement is not abolished, but it is postponed to a separate step that comes later and is conducted differently.

The distinction matters in practice because teams slide from generating into choosing without noticing. The symptom is easy to recognise: as soon as people begin arguing about whether an idea will work, the session has stopped being ideation and has become convergence – usually convergence on the first idea anybody offered.

Ideation ends with a wide field of options, not with a decision. The moment the room starts defending a single idea, the divergent half is over.

2. Quantity before quality

The governing principle of the divergent half is that quantity precedes quality. The first ten ideas produced by any group are the obvious ones – everybody in the room already had them before the session began, and they are usually the ideas that have already been tried. Unusual ideas appear only after the obvious ones have been said out loud and cleared away.

- A weak idea is frequently the raw material for a strong one belonging to somebody else; discarding it early removes that possibility.
- Because nothing is being committed to, a bad idea costs nothing. The only real cost in a divergent session is time.
- Set a numerical target and a clock, and do not stop when the room first goes quiet – the silence after the obvious ideas run out is where the session begins to earn its keep.

KEY POINT Aim for a hundred ideas in an hour. A team that produces twelve ideas and then discusses them has not held an ideation session; it has held a meeting.

3. A map of the techniques and three shapes of session

The techniques in this unit differ along two axes: whether ideas are produced by **speaking** or by **writing**, and whether the session is **free** or **structured**. Placing them on those axes shows why a team should not use only one.

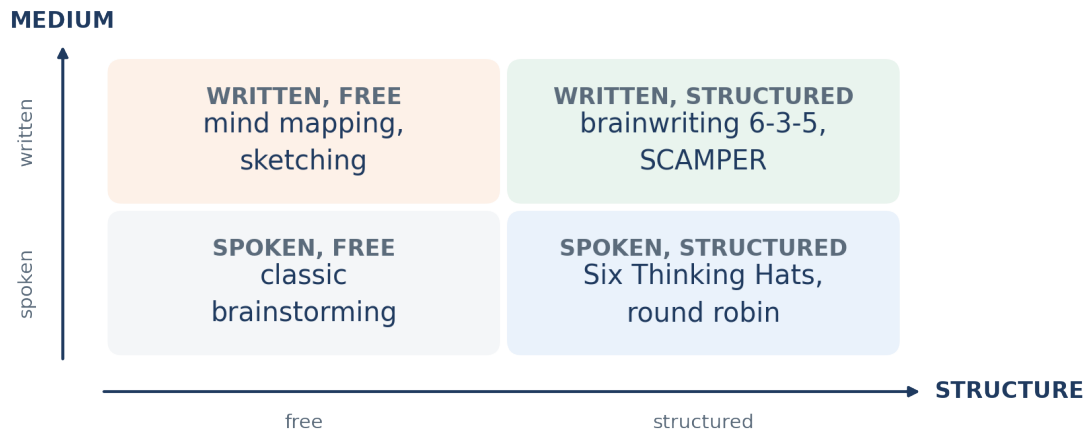


Figure 3.1 The techniques of this unit placed by medium and by degree of structure. All four quadrants are visited in the six hours.

Shape of session	What happens	Strength and weakness
Solo first	Everyone writes alone before anyone speaks	Protects quiet members and unusual ideas; loses the energy of a group
Group together	Ideas are built on out loud, in one conversation	Fast and social; easily dominated by the most confident voice
Solo, then group	Write alone, then share and build together	The shape most sessions should use; needs a facilitator to enforce

KEY POINT Start alone and finish together. Starting together loses every idea that was not spoken first, because the room anchors on whatever was said at the beginning.

4. Running an ideation session

Most of the quality of a session is determined before it begins. A session with a vague question, the wrong people and no materials cannot be rescued by enthusiasm.

Stage	What it requires
Before	One framed question, written large and visible to everyone; six to eight people including at least one non-engineer; markers, sticky notes, wall space and a visible clock; a facilitator who has agreed not to contribute ideas
During	A five-minute warm-up on something unrelated; the rules read aloud and agreed; one idea per note, written large enough to read from three metres; a new angle called whenever the flow slows – a constraint, an analogy, a reversal
After	Photograph the wall before anything is moved; cluster as a group and name each cluster in a sentence; mark the wild ideas and carry at least one into every later round; agree the next step and its owner before the room empties

Four failures account for most unproductive sessions. The senior person speaks first, so the room anchors within two minutes. Judgement arrives early, and after one “that will not work” the unusual ideas stop being offered at all. One person talks throughout, which is **production blocking**: while one person speaks, five others are not writing. And the question has no constraint at all, so the ideas are vague and nobody can build on them.

The facilitator does	The facilitator does not
Keep time out loud	Contribute ideas of their own
Enforce the agreed rules	Rank or judge anything during the session
Write nothing themselves	Let one voice dominate the room
Call a new angle when flow dies	Let the session run past the clock

KEY POINT A facilitator who joins in stops being a facilitator, and the room loses its referee at exactly the moment it needs one.

IN CLASS Forty-five seconds each, round the room: name one use for a brick. No repeats, no judging, no pauses. The exercise is a warm-up, and it also demonstrates fluency and flexibility in under a minute.

5. Brainstorming – origins and rules

Brainstorming is the oldest formal ideation method and the one most often done badly. Alex Osborn, an advertising executive, developed it from 1939 and published it in *Applied Imagination* in 1953. His four original rules were: go for quantity; withhold criticism; welcome wild ideas; and combine and improve the ideas of others. He summarised the fourth as “1 + 1 = 3” – ideas breed on one another. The Stanford d.school later added three rules governing how the room itself behaves.

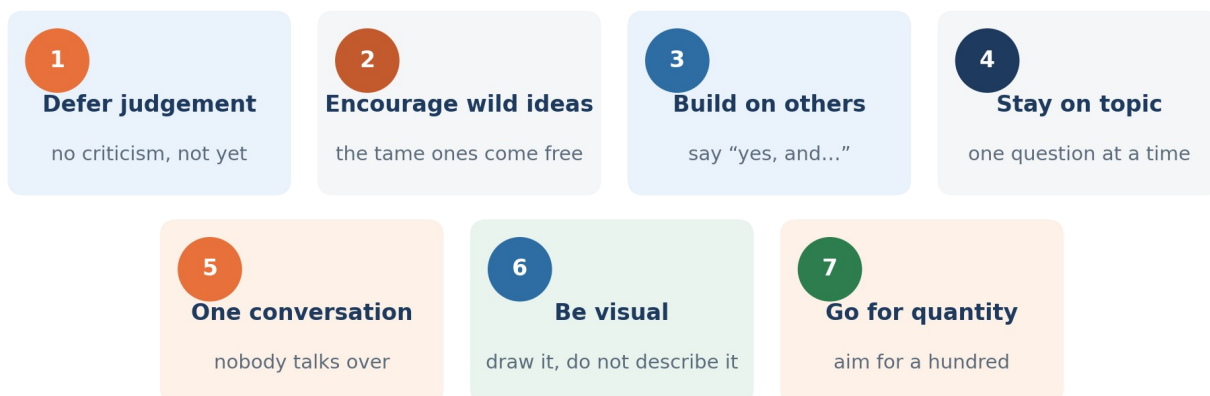


Figure 3.2 The seven rules as commonly taught. They are read aloud at the start of a session; a rule nobody has agreed to is not a rule.

Rule	The failure it prevents
Defer judgement	Ideas that sound wrong but are not being killed on sight
Encourage wild ideas	A narrow field in which every idea is a variation of one idea

Rule	The failure it prevents
Build on others	A group producing no more than the same people working alone
Stay on topic	A session that drifts and produces nothing usable
One conversation at a time	Half the room being unable to hear or contribute
Be visual	Ideas too vague for anybody else to build on
Go for quantity	Stopping at the obvious answers

6. Why brainstorming often underperforms

The research evidence on group brainstorming is uncomfortable and worth knowing. Michael Diehl and Wolfgang Stroebe, reviewing twenty-two studies in 1987, found that groups brainstorming together produce **fewer** ideas than the same number of people working separately and pooling their output. The finding has been replicated many times. Three mechanisms explain it.

Cause	What happens	The remedy
Production blocking	Only one person can speak at a time, so the others wait instead of thinking	Write rather than speak
Evaluation apprehension	People hold back whatever might make them look foolish	Make the output anonymous or simultaneous
Anchoring	The whole room circles the first idea that was offered	Generate alone before anybody speaks

KEY POINT The problem is not the rules; it is the talking. Every remedy in the right-hand column above is a way of removing the queue – which is precisely what brainwriting does.

7. Brainwriting 6-3-5 and other written methods

Bernd Rohrbach published the 6-3-5 method in 1968 in the German journal *Absatzwirtschaft*. The numbers name the mechanism: **six** participants, **three** ideas each, **five** minutes a round. Each person writes three ideas on a row of a worksheet and passes the sheet on; the next person reads what is there and adds three more, either extending the earlier ideas or reacting against them. Six rounds take thirty minutes and produce 108 ideas, and not a word is spoken.

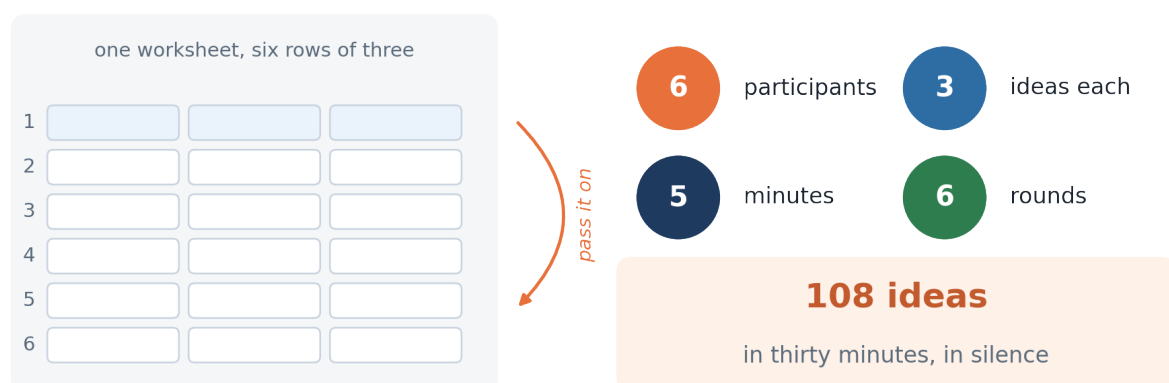


Figure 3.3 The 6-3-5 worksheet and its arithmetic. The silence is the point, not an incidental feature.

	Brainstorming	Brainwriting
Medium	Spoken	Written
Turn taking	One person at a time	Everyone at once
Quiet participants	Frequently lost	Equally represented
The record	Needs a scribe	Records itself, with authorship
Energy in the room	High and social	Low and focused

Three other written methods do the same job of removing the queue. In the **brainwriting pool**, sheets go into the middle of the table; a participant takes any sheet, adds to it and returns it. **Crazy Eights** folds a sheet into eight boxes and asks for eight sketches in eight minutes, one a minute, which forces quantity and forces drawing. **Round robin** takes one idea from each person in turn until somebody passes twice.

KEY POINT Most good sessions use both media: brainwriting to generate the field in silence, then brainstorming to build on it out loud. The order matters.

IN CLASS Thirty minutes, in silence. Write your team's How Might We question at the top of a sheet; six rows of three boxes, one sheet per person; five minutes a round, then pass left, reading before writing. After six rounds, count the ideas and mark the five most unexpected.

8. Mind mapping

A mind map is a radial diagram: one central idea, main branches radiating from it, and sub-branches off those. Tony Buzan popularised the form through the BBC series *Use Your Head* in 1974, although diagrams of this kind are far older. The essential features are that entries are **keywords** rather than sentences, and that colour and position are structural rather than decorative – they keep the branches distinguishable.

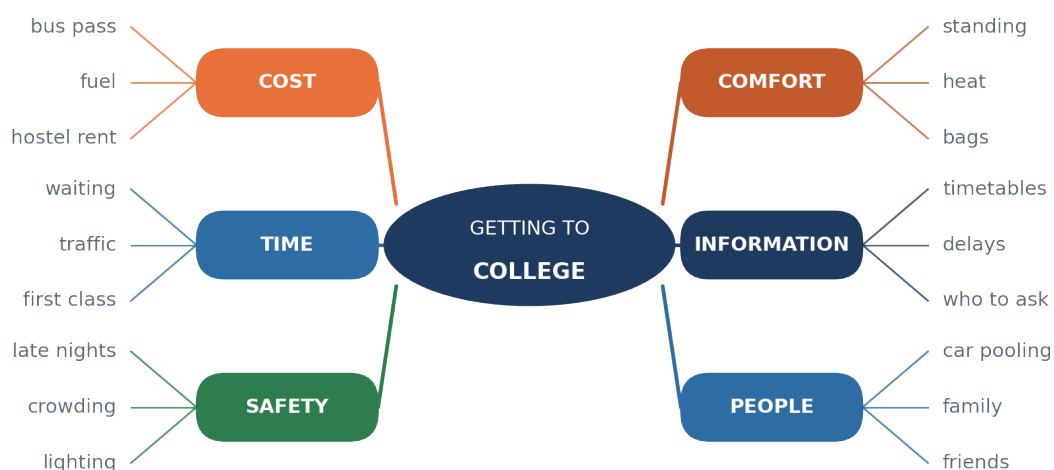


Figure 3.4 A worked mind map of a student problem. The short branches are the interesting ones.

- Turn the page landscape and start in the centre with the topic.
- Keep the main branches to five or seven; more than that and the map stops being readable.
- One keyword to a branch, and add sub-branches the moment they occur to you.
- Work fast and do not correct. A messy first map is doing its job; a beautiful one usually means the author stopped generating and started tidying.

	Mind map	Concept map
Structure	Radial tree from one central topic	A network of several concepts
Links	Unlabelled branches	Labelled with verbs or relations
Purpose	Opening a topic up	Explaining how a system works
Used in design for	Unpacking a problem, generating, clustering	Communicating a finished understanding

Reading a finished map is a skill in itself. Branches that grew long show where the team's thinking already was; branches that stayed short show where nobody has looked. Circle the crossings – the same idea appearing on two branches is a theme – and name each theme in a sentence to carry forward as a How Might We question.

KEY POINT The short branches are the interesting ones. They mark the parts of the problem that nobody in the team has thought about yet, which is exactly where the unclaimed ideas are.

9. The SCAMPER technique

SCAMPER is a forced-association tool: rather than waiting for inspiration, it supplies seven prompts and requires an answer to each. Alex Osborn published a checklist of idea-spurring questions in *Applied Imagination* in 1953, and Bob Eberle turned that checklist into the SCAMPER mnemonic in 1971. It works best on something that already exists and needs improving.



Figure 3.5 The seven SCAMPER prompts. Every letter is answered, including the ones that seem to lead nowhere.

Letter	What it asks	Applied to a campus water cooler
Substitute	What else instead?	A clay pot instead of a plastic tank
Combine	What could be joined?	Cooler plus a bottle-refill counter, one queue
Adapt	What is like this?	Borrow the token system used at a bank counter
Modify	Bigger, smaller, other?	Many small points instead of one large one
Put to use	Who else could use it?	Open it to the road-facing gate
Eliminate	What can be removed?	Remove the cup; everyone carries a bottle

Letter	What it asks	Applied to a campus water cooler
Reverse	What if it ran backwards?	The water comes to the corridor

- Name the object or process precisely – “the queue at the water cooler”, not “water”.
- Take one letter at a time and give it a fixed three minutes.
- Write every answer, including the obviously impractical ones; they are usually one step from a practical one and nobody else in the room would have written them.
- Evaluate nothing until all seven letters are done.

KEY POINT SCAMPER improves something that exists. It will not tell you what the problem is – that is what Unit II was for – and it is useless on a blank page.

10. Lateral thinking

The techniques so far have all worked by making a group more productive. Lateral thinking, a term introduced by **Edward de Bono** in 1967, works on something different: it attacks the shape of the thinking itself. De Bono's argument is that the mind is a self-organising system. Incoming information settles into patterns, and once a pattern exists, later information runs into it as rain runs into an existing channel. This is enormously useful — it is what allows you to recognise a face or read a word without effort — but it means that a familiar problem is almost always thought about in the familiar way.

Vertical thinking digs the existing hole deeper. It is selective, sequential and correct at every step: each move follows from the last, and a wrong move is discarded at once. It is the thinking of analysis, of calculation and of examinations, and engineering education trains it well. **Lateral thinking** digs a hole somewhere else. It is generative rather than selective, it may move sideways or backwards, and — this is the part students find hardest — it allows a step to be wrong on the way to being useful. The two are not rivals. A concept that lateral thinking produces must still be made to work by vertical thinking; lateral thinking without follow-through produces only amusement.

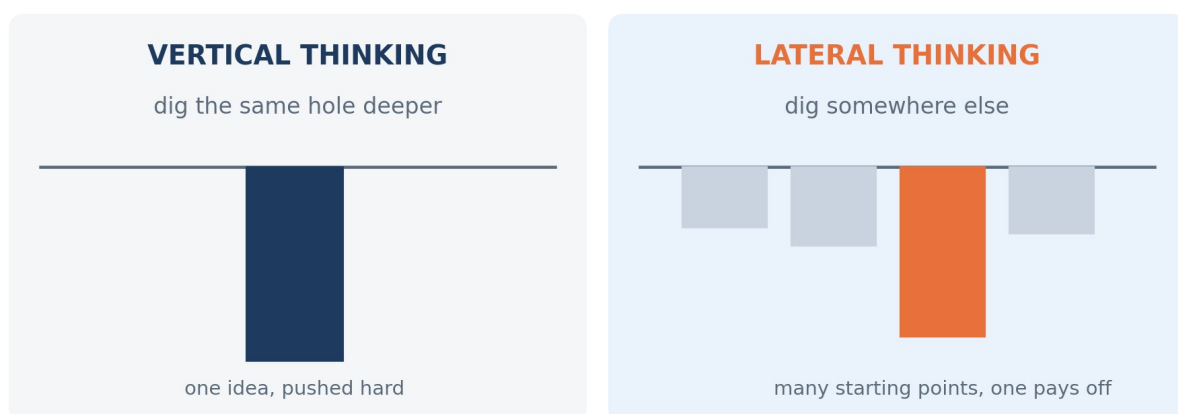


Figure 3.6 Vertical thinking deepens one line of attack; lateral thinking moves across to start a new one. Design work needs both, in that order.

10.1 Provocation and movement

The central device of lateral thinking is the **provocation** — a deliberately unreasonable statement made not because it is true but because of where thinking has to go to deal with it. De Bono coined the word **po** to mark such a statement, signalling to everyone present that what follows is a stepping stone and

not a proposal. *Po: the bus has no driver. Po: the hostel mess serves no food. Po: the examination has no paper.* None of these is a suggestion. Each is a place to stand.

What is then required is **movement**, which is a different mental operation from judgement. Judgement asks whether the statement is true; movement asks what follows from it. There are four standard ways to extract movement from a provocation, and a class that is taught them will use a provocation productively instead of laughing at it.

Way to move	What you do	Applied to “Po: the bus has no driver”
Extract the principle	Name the principle in the provocation and keep only that	The principle is “no one person controls the vehicle” — passengers could control the stopping
Focus on the difference	Ask what is different about this world and what that difference buys	No wage cost and no driver fatigue — so a night service becomes affordable
Moment to moment	Run the film forward and describe what happens next	Passengers board with no one watching — so fare collection must move to the stop
Positive aspects	Name what is good in the provocation even if the whole is absurd	No argument with a conductor, no favouritism at stops

10.2 Random entry

The second standard tool is **random entry**. A word is taken at random — from a dictionary, from a prepared list of sixty nouns, or from whatever the person sitting to your left can see — and the team is required to connect it to the problem. Because the word is genuinely unrelated, it enters the mind at a point far from the worn channel, and the route back to the problem passes through territory that would never have been visited. The rule is that the first word must be used: a team allowed to reject words will quietly select the word nearest its existing idea, and the technique collapses.

A random word works because it is random. The moment you choose a suitable word, you have chosen your existing answer.

An example. The problem is queueing at the campus photocopier; the random word is **umbrella**. An umbrella is carried but used rarely; it is shared in a rain shower; it is left behind in shops; it opens only when needed. Following “used rarely but carried always” gives the idea of a copier account on the student ID rather than a cash queue. Following “left behind in shops” gives a collection shelf for finished jobs so that the machine is never blocked by a person waiting to collect. Neither idea is about umbrellas.

10.3 Other lateral moves

- **Challenge.** Ask why something is done the way it is done — not because it is wrong, but to see whether the reason still holds. Most answers of the form “because it has always been so” mark a free design opportunity.
- **Concept fan.** Move up from the specific idea to the concept it serves, and from the concept to the direction, then come back down a different branch. Idea: add more counters. Concept: reduce waiting. Direction: change the demand rather than the supply.
- **Reversal.** State the opposite of the usual relation and see what it yields. Instead of the student going to the office, the office goes to the student; instead of the library lending books, readers lend books to the library.

- **Escape.** List what is taken for granted, then remove one item. A hospital assumes the patient travels to it; remove that and you have a mobile clinic. A form assumes a literate applicant; remove that and you have a voice interface.

KEY POINT Lateral thinking is not the same as “being creative” and it is not a personality trait. It is a set of deliberate operations — provocation, movement, random entry, challenge, reversal, escape — that can be taught, practised and timed. Anyone can be walked through them in ten minutes.

11. The Six Thinking Hats

The second de Bono method in the syllabus addresses a different failure. In an ordinary discussion everyone is doing everything at once: one member is enthusiastic, another is listing risks, a third is asking for data and a fourth is proposing an alternative — and because they are doing different kinds of thinking simultaneously, they appear to be disagreeing when often they are not. Worse, positions harden: the person who raised the first objection becomes the critic for the rest of the meeting and must defend that role.

Six Thinking Hats, published in 1985, replaces this with **parallel thinking**. At any moment the whole group wears the same notional hat, which means everyone is doing the same kind of thinking at the same time, and then the group changes hat together. Nobody owns the criticism, because everyone criticises during the black hat and nobody criticises outside it. The hats are a convention for directing attention, not a description of personality; the sentence that matters in a session is “that is a black-hat point, hold it for two minutes” rather than “you are being negative”.

Everyone wears the same hat at the same time



parallel thinking: the team looks in one direction, then turns together

Figure 3.7 The six hats. Blue directs the session; the other five are modes of thinking that the whole group enters and leaves together.

Hat	Mode of thinking	Typical sentence
White	Information: what we know, what we need, where the gap is	“We have footfall for Monday only; we do not have weekend data.”
Red	Feeling: intuition and emotion, stated without justification	“I dislike the second concept. I cannot say why.”
Black	Caution: risks, faults, why it may not work — logical, not hostile	“The charging point fails if the power cut lasts past four hours.”
Yellow	Value: benefits, feasibility, why it may work	“If it works, the same rack serves the hostel and the library.”
Green	Growth: new ideas, alternatives, modifications	“Suppose the rack were mounted on the bus instead.”

Hat	Mode of thinking	Typical sentence
Blue	Process: agenda, sequence, summary, next step	"Three minutes of yellow, then we decide."

11.1 Running a hats session

The facilitator wears blue at the start and at the end, and states the sequence before the discussion begins. A short standard sequence for evaluating a concept is blue, white, green, yellow, black, red, blue — information first so that opinion rests on something, possibility before criticism so that a fragile idea is not killed at birth, and feeling last so that it is recorded rather than disguised as argument. Each hat is timed, typically two to four minutes; the discipline comes from the clock as much as from the rule. Red is deliberately kept to about thirty seconds a person, because a feeling stated briefly is data and a feeling defended at length is a quarrel.

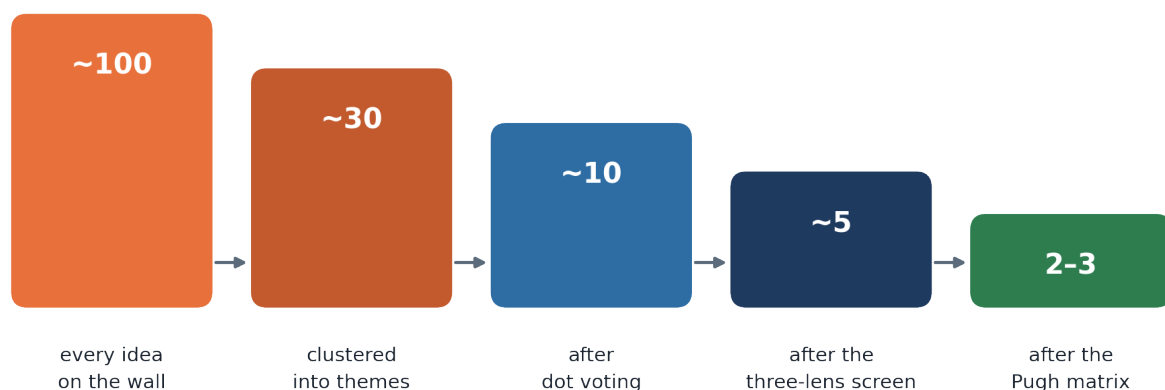
Two rules make the difference between a hats session and a normal meeting. First, **no hat may be worn out of turn**; a black-hat point raised during the yellow hat is written on the board and taken up when the black hat comes. Second, **everyone wears every hat**; the optimist must find risks and the sceptic must find benefits, which is precisely the value of the method for a team that has settled into fixed roles.

KEY POINT The black hat is the most useful hat and the most abused. It is for logical caution supported by a reason, not for dislike; "this will not work because the mess closes at nine" is black-hat thinking, "this is a silly idea" is not.

IN CLASS Thirty minutes. Take the two strongest concepts from your team's ideation session. Run a full hats sequence on each — blue, white, green, yellow, black, red, blue — with two minutes a hat and one member keeping time and wearing blue. Record the output of each hat on a separate sheet. Compare how the two concepts look after the session with how they looked before it.

12. Converging – clustering, dot voting and quick screens

Divergence without convergence is a wall of paper. A team that has produced a hundred and fifty ideas must now reduce them to two or three that will be prototyped, and it must do so in a way that is quick, visible and defensible. The shape of the whole unit is therefore a funnel: many ideas, then a few themes, then a shortlist, then the concepts that survive a structured comparison.



keep one wild card through every screen — the safe list is rarely the interesting one

Figure 3.8 Convergence in stages. Each stage uses a cheaper method than the one after it, so that expensive analysis is spent only on survivors.

12.1 Clustering and naming

The first move is the same synthesis operation met in Unit II. Ideas are grouped by the approach they take rather than by the wording on the note, and each group is given a name that is a short phrase rather than a single word: *“bring the service to the user”* rather than *“logistics”*. Ten to fifteen named clusters usually emerge from a hundred and fifty ideas, and the naming matters because the team from here on argues about approaches, of which there are a dozen, rather than about ideas, of which there are too many to hold in mind.

12.2 Dot voting

Each member is given a fixed number of adhesive dots — three or four is usual — and places them on the ideas or clusters they consider strongest. Votes are placed simultaneously and silently, which is the whole point: sequential or spoken voting produces a cascade in which later voters follow earlier ones. Members may put more than one dot on a single idea if they feel strongly. The result is a rough preference ordering obtained in four minutes.

- **Vote after clustering, not before.** Voting on raw notes rewards whichever idea happened to be written most legibly.
- **Vote silently and at once.** If the team leader's dot goes up first, the exercise measures deference.
- **Use coloured dots for different criteria.** Green for “most likely to delight the user”, yellow for “most likely to be buildable in three weeks”; the two patterns rarely coincide and the difference is informative.
- **Treat the result as data, not as a decision.** Dot voting shows where enthusiasm is. It does not show which concept is best, and a single dot on a strange idea may be the most interesting mark on the wall.

12.3 Deliberate selection across categories

A team left to itself will shortlist three safe ideas, because safe ideas attract dots. A useful corrective is to require the shortlist to contain one idea from each of several categories fixed in advance — for example **the rational choice** (clearly sensible and buildable), **the delight** (the one the team most wants to see exist), **the darling** (the one a member is attached to and cannot let go) and **the long shot** (the one that would change everything if it worked). Carrying a long shot into prototyping costs very little at this stage, and it is the category that produces the surprises.

KEY POINT Convergence is a decision, not a discovery. Nothing in the pile tells you which idea is best; the team must choose, and the value of a method is that it makes the basis of the choice visible to everyone afterwards.

13. The three-lens screen and the Pugh matrix

13.1 Desirability, feasibility, viability

The standard first screen applies three lenses to each shortlisted concept. **Desirability** asks whether people want it — whether it answers a need actually observed in the field rather than one assumed at the desk. **Feasibility** asks whether it can be built with the technology, skill and time available. **Viability** asks whether it can be sustained — whether the money, the staffing or the maintenance work over the life of the solution. A concept must eventually sit in the intersection of all three; a concept strong on only one lens is, respectively, a wish, a gadget or a spreadsheet.

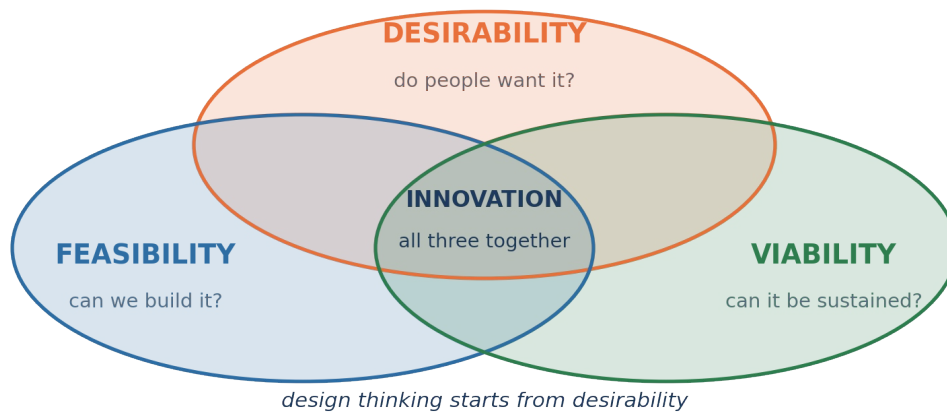


Figure 3.9 The three lenses. Innovation lives in the intersection; each pair without the third has a familiar failure mode.

Two cautions are worth stating to a first-year class. First, the lenses are applied **in that order**: a concept nobody wants is not rescued by being cheap to build. Second, feasibility and viability are properties of a concept **at a point in time**. A concept judged infeasible in week two may become feasible after a prototype teaches the team something, so the screen records a reason rather than a verdict — “infeasible because we cannot source a food-safe coating” is a task, whereas “infeasible” is a dead end.

13.2 The Pugh matrix

For the final comparison the method of choice is the **Pugh matrix**, also called the decision matrix or controlled convergence, introduced by **Stuart Pugh** of the University of Strathclyde in the 1980s. Its distinguishing feature is that concepts are not scored in the abstract but compared against a **datum** — an existing solution or the strongest current concept — using nothing more than better, same or worse.

1. List the criteria down the left, drawn from the user needs and constraints established in Unit II, not invented at the matrix.
2. Choose a datum: the present way of doing things, or the concept the team currently favours. Put it in the first column.
3. Score every other concept against the datum on each criterion: + for better, – for worse, S for the same. Judge one criterion at a time across all concepts, not one concept at a time down all criteria.
4. Total the pluses, the minuses and the same separately. Do not net them into a single number at this stage; a concept with six pluses and five minuses is a different animal from one with one plus and no minuses.
5. Read the matrix for insight, not for a winner. A column of minuses on one criterion means the criterion may be wrong or the whole set of concepts is weak there.
6. Run a second round. Strengthen a promising concept by borrowing the feature that earned a plus elsewhere, change the datum to the leading concept, and score again. Pugh called this controlled convergence, and the second round is where the method earns its keep.

Criterion	Datum: present hostel notice board	C1: mobile alerts	C2: mess-door screen	C3: class rep relay
Reaches every resident	Datum	+	S	–
Works without a smartphone	Datum	–	+	+
Message cannot be missed	Datum	+	+	S
Cost to set up	Datum	S	–	+

Criterion	Datum: present hostel notice board	C1: mobile alerts	C2: mess-door screen	C3: class rep relay
Effort to maintain daily	Datum	+	S	–
Total + / – / S	—	3 / 1 / 1	2 / 1 / 2	2 / 2 / 1

The matrix above is read as follows. C1 leads, but its single minus is on a criterion the team cannot compromise, so the second round asks how C1 might reach residents without a smartphone — and the answer, borrowed from C2, is a screen at the mess door carrying the same feed. That hybrid is a concept that existed in no column, and producing it is exactly what the method is for.

KEY POINT A Pugh matrix is not arithmetic. The numbers at the foot of the columns are a summary of a conversation, and the conversation about why a cell holds a minus is worth more than the total beneath it.

13.3 Weighted matrices, and when not to weight

Criteria are rarely equal, and a weighted matrix multiplies each score by a weight agreed in advance. This is appropriate when the criteria genuinely differ in importance and the team can defend the weights. It is inappropriate — and dangerous — when weights are assigned after the scores are visible, because weights can be tuned until the favoured concept wins, and the result then carries the authority of a calculation while containing none of its discipline. If weights are used, fix them and write them down before any concept is scored.

IN CLASS Thirty-five minutes. Shortlist four concepts from your clusters, choosing one rational, one delight, one darling and one long shot. Screen all four on desirability, feasibility and viability, recording a reason in every cell. Then build a Pugh matrix with at least six criteria taken from your Unit II problem statement, run a second round with the leader as datum, and record which hybrid concept the second round produced.

14. Traps in concept selection

Selection is where a good ideation session is most often wasted. The traps below are common enough that a team should read the list aloud before it starts to converge.

Trap	What it looks like	What to do instead
Choosing before generating	The team knew the answer in week one and the session decorates it	Make the datum the week-one idea; anything that cannot beat it is not worth building
Averaging	Features of five concepts are merged until nothing is distinctive	Hybridise deliberately by borrowing one strength, not by taking the mean
Criteria invented at the matrix	Criteria appear that no user ever mentioned	Draw criteria from the field notes and the problem statement
False precision	Scores of 7.5 out of 10 on judgements nobody can defend	Use +, – and S until you have data that earns a number
The loudest voice	The concept belonging to the most confident member survives	Score silently first, then discuss the cells that disagree

Trap	What it looks like	What to do instead
Killing the long shot early	Only safe concepts reach prototyping, and nothing is learnt	Reserve one shortlist slot for the concept that would change everything
Selecting on cost too early	Costs are guesses at this stage, and the cheapest concept wins by default	Screen on desirability first; cost decides between concepts users want

The purpose of selection is not to be right. It is to choose, on visible grounds, what the team will build next — and to be able to change its mind when the prototype teaches it something.

15. Summary

- Ideation converts one well-framed problem into many possible answers; its output is raw material for prototyping, not a decision, and it is worthless without the problem statement that precedes it.
- Quantity is pursued before quality because the first ideas out of a group are the obvious ones; the unusual answers arrive after the obvious ones are exhausted.
- Techniques differ by medium (speaking or writing) and by structure (free or constrained); a team that uses only one quadrant gets only one kind of idea.
- A session needs a framed question, a facilitator, a visible timer, materials that make ideas public, and a rule that judgement is suspended until convergence begins.
- Osborn's four rules are defer judgement, welcome wild ideas, seek quantity and build on the ideas of others; the rules exist because each names a specific way groups destroy their own output.
- Interacting groups underperform nominal groups because of production blocking, evaluation apprehension, free riding and anchoring — all four are structural, not personal.
- Brainwriting 6-3-5 removes production blocking and evaluation apprehension by making everyone write at once; six members, three ideas, five minutes, six rounds yields 108 ideas in half an hour.
- Mind mapping organises a single mind's associations radially from a central problem; branches are named with single words and the map is read for clusters.
- SCAMPER supplies seven forced prompts — substitute, combine, adapt, modify, put to another use, eliminate, reverse — and is the fastest way to improve an existing product rather than invent a new one.
- Lateral thinking attacks the pattern itself using provocation with movement, random entry, challenge, reversal and escape; vertical thinking then makes the result work.
- The Six Thinking Hats replaces argument with parallel thinking: the whole group wears one hat at a time, so nobody owns the criticism and fixed roles dissolve.
- Convergence proceeds in stages — cluster and name, dot vote silently, shortlist across fixed categories — so that expensive analysis is spent only on survivors.
- The three lenses are desirability, feasibility and viability, applied in that order; a screen records the reason for a verdict, because reasons become tasks.
- A Pugh matrix compares concepts against a datum using +, – and S, and its value lies in the second round, where a hybrid concept that appeared in no column emerges.
- The traps are choosing before generating, averaging, inventing criteria at the matrix, false precision, the loudest voice, killing the long shot and screening on cost too early.

16. Review questions

16.1 Short-answer (two-mark) questions

1. State the purpose of ideation in one sentence and say what its output is.
2. Why is quantity sought before quality in idea generation?
3. Name the two axes on which ideation techniques may be mapped.
4. List four things a facilitator must arrange before a session begins.
5. State Osborn's four rules of brainstorming.
6. What is production blocking? Which technique removes it?
7. Distinguish a nominal group from an interacting group.
8. Define evaluation apprehension and give one way of reducing it.
9. Explain the numbers in the name 6-3-5 and state the total idea count.
10. Name two written ideation methods other than 6-3-5.
11. State three rules for drawing a mind map.
12. Expand SCAMPER.
13. Apply any two SCAMPER prompts to a ceiling fan.
14. Distinguish vertical thinking from lateral thinking.
15. What is a provocation? What does the marker "po" signify?
16. Name the four ways of obtaining movement from a provocation.
17. Why must the first random word be used rather than a chosen one?
18. Name the six thinking hats and the mode of thinking each directs.
19. Why is the red hat kept short?
20. State two rules that distinguish a hats session from an ordinary meeting.
21. Why is dot voting done silently and simultaneously?
22. Name the four categories a balanced shortlist should contain.
23. State the three lenses of the standard concept screen.
24. What is a datum in a Pugh matrix?
25. Why are the pluses and minuses not netted into a single number?
26. When is a weighted decision matrix inappropriate?

16.2 Long-answer (thirteen-mark) questions

1. Explain the purpose of ideation and the principle of quantity before quality. Describe the conditions that must be arranged before a session and the role of the facilitator during it.
2. State and explain Osborn's four rules of brainstorming. For each rule, identify the specific group behaviour it is designed to prevent, with an example from a student team.
3. Discuss the four causes of productivity loss in interacting groups. Explain how each is addressed by a structural change to the session rather than by exhorting members to behave differently.
4. Describe the 6-3-5 brainwriting method in full, including the sheet layout, the timing and the rotation. Compare its output and its failure modes with those of oral brainstorming.
5. Explain mind mapping. Describe how a map is constructed, the rules that keep it readable, and how a finished map is converted into named themes for a team.
6. Explain all seven SCAMPER prompts. Apply the complete set to a single familiar product of your choice, producing at least two distinct ideas per prompt.
7. Distinguish vertical from lateral thinking with examples. Explain provocation and the four ways of obtaining movement, illustrating with a provocation of your own.

8. Describe random entry, challenge, reversal and escape as lateral-thinking operations. Work one problem through each of the four and compare the ideas they produce.
9. Explain the Six Thinking Hats method and the principle of parallel thinking. Describe a sequence suitable for evaluating a concept and justify the order you choose.
10. Describe the convergence funnel from a hundred and fifty ideas to three concepts. Explain clustering, dot voting and deliberate selection across categories, and state the danger each stage is guarding against.
11. Explain desirability, feasibility and viability as screening lenses. Discuss the characteristic failure of a concept that satisfies only two of the three, with an example of each of the three pairs.
12. Explain the Pugh matrix in full: criteria, datum, scoring, totals and the second round. Construct a worked matrix of at least five criteria and four concepts and interpret the result.
13. Discuss the common traps in concept selection. For each, describe how it arises in a student team and state the practice that prevents it.

Text: Tim Brown, *Change by Design*, Harper Collins, 2012 — the chapters on convergent and divergent thinking and on building to think; Idris Mootee, *Design Thinking for Strategic Innovation*, Wiley, 2013 — Chapters 5 and 6. **References:** Alex F. Osborn, *Applied Imagination: Principles and Procedures of Creative Thinking*, Scribner, 1953; Bernd Rohrbach, "Kreativ nach Regeln — Methode 635", *Absatzwirtschaft*, 1969; Bob Eberle, *SCAMPER: Games for Imagination Development*, 1971; Edward de Bono, *Lateral Thinking*, Ward Lock, 1970, and *Six Thinking Hats*, Little Brown, 1985; Tony Buzan, *The Mind Map Book*, BBC Books, 1993; Stuart Pugh, *Total Design*, Addison-Wesley, 1990; Nigel Cross, *Engineering Design Methods*, 4th Edn., Wiley, 2008 — the chapters on generating alternatives and evaluating alternatives; Michael Diehl and Wolfgang Stroebe, "Productivity Loss in Brainstorming Groups", *Journal of Personality and Social Psychology*, 1987.
